

SUMMARY

Staff-level platform and distributed-systems engineer with 12+ years designing, building, and operating foundational software across Kubernetes, cloud infrastructure, Linux, and developer platforms. I lead ambiguous, cross-functional initiatives from architecture through production, with a focus on reliable service infrastructure, cloud-native systems, security, and developer experience. I build platforms and AI-enabled automation that reduce operational toil, improve reliability and security, and help engineering teams ship faster and more safely.

SKILLS

Languages: Go, Python, Bash

Distributed Systems & Platform: Kubernetes, Linux, Helm, ArgoCD, Flux, Backstage, service-oriented systems

Cloud & Infrastructure: AWS, GCP, Terraform, Ansible, Packer, multi-account and multi-region deployments

Observability & Reliability: Prometheus, Grafana, OpenTelemetry, ELK

Security & Delivery: SPIRE, OPA, Vault, mTLS, Jenkins, GitHub Actions, Semaphore, Atlantis

EXPERIENCE

- **Confluent** Remote
Senior Software Engineer II, Developer Productivity *April 2025 - Present*
 - **Production service architecture:** Re-architected a production policy-enforcement service that silently dropped critical configuration-change approvals when synchronous processing exceeded an upstream timeout. Designed an asynchronous, back-pressured architecture with operational dashboards and alerting, eliminating the silent-failure mode and the on-call toil it caused.
 - **Developer platform:** Co-leading technical direction for the internal developer platform, owning the developer-experience and self-service roadmap and shaping foundational capabilities that help engineering teams provision, deploy, and operate services more reliably.
 - **AI gateway architecture:** Leading evaluation and architecture of the company's AI gateway, a shared control plane for how services access large language models. Drove cross-team technical due diligence and source-level verification that caught a supply-chain compromise in a leading candidate before adoption; an earlier AI-infrastructure evaluation I led avoided more than \$500K in annual spend by recommending against an unready deployment.
 - **Autonomous vulnerability remediation:** Designed and shipped an AI agent that automates the end-to-end software-vulnerability remediation lifecycle; adopted as the organization's default, it remediated 300+ vulnerabilities in roughly five months. Championed its migration to a new autonomous-workflow platform and drove cross-team design reviews for the next architecture.
 - **Zero-trust service infrastructure:** Drove fleet-wide security and reliability hardening by automating service migration to zero-trust mTLS and modern least-privilege secret management, reducing operational toil while shrinking the credential attack surface.
 - **Cross-org technical leadership:** Authored RFCs and technical one-pagers that steered cross-team platform and AI-tooling decisions; taught engineers across the organization to build high-quality AI-assistant plugins and published reusable plugins to the company's internal marketplace.
- **Angi Inc** Remote
Senior Infrastructure Engineer *Sept 2023 - Mar 2025*
 - **Cloud-native developer platform:** Built foundational components of the Angi One internal platform using Kubernetes controllers, Flux, and ArgoCD, enabling developers to provision applications and cloud services from a single declarative manifest and cutting production delivery time by 50%.
 - **Platform product experience:** Partnered with application and observability teams to automate Backstage catalog component generation and improve service onboarding and system visibility across the engineering organization.
 - **Developer tooling:** Led automation of documentation workflows from Kubernetes CRDs so platform documentation evolved with releases, reducing manual upkeep and support overhead.
- **Workday** Victoria, BC, Canada
Senior Software Development Engineer, DevOps *May 2023 - Sept 2023*
Software Development Engineer III, DevOps *May 2021 - April 2023*
Software Development Engineer II, DevOps *May 2019 - April 2021*

- **Distributed deployment platform:** Designed an automated AWS deployment pipeline for microservices across accounts and regions, standardizing service delivery and reducing recurring operations toil by more than four hours per week.
- **Policy and self-service infrastructure:** Implemented policy-based infrastructure workflows using OPA and Atlantis, giving developers safer self-service control over cloud resources while shifting security checks earlier in the delivery lifecycle.
- **Cloud and Kubernetes platform engineering:** Built and operated production infrastructure across AWS and datacenter environments, including EC2, ECS, Route53, S3, RDS, Lambda, and Elasticsearch, while migrating applications and services onto an internal Kubernetes platform.
- **CI/CD platform:** Developed Jenkins pipelines and shared delivery libraries used by multiple applications and services, improving consistency and repeatability across build and deployment workflows.
- **Technical leadership:** Spearheaded a cross-team enablement initiative that helped development teams take greater ownership of operating their services; mentored engineers as they grew into larger technical responsibilities.

- **NYU CUSP**

Associate Research Scientist
Assistant Research Scientist

Brooklyn, NY, USA
 May 2015 - May 2019
 June 2014 - May 2015

- **Container orchestration:** Developed Dockkeeper, a scalable and secure container scheduling and monitoring system using Docker and Prometheus to provision services across physical hosts; eliminated more than \$35K per year in VM licensing costs and improved host utilization by over 55%.
- **Distributed compute and storage infrastructure:** Architected the NYU/CUSP Urban Observatory's multi-site physical infrastructure with dense compute and more than half a petabyte of storage, provisioned as multi-user mini-HPC environments using Ansible and Packer.
- **Production IoT platform:** Developed a secure, mission-critical IoT platform and CI/CD framework for deploying and operating more than 100 urban noise-monitoring sensors across New York City as part of the NSF-funded SONYC project.

EDUCATION

- **NYU Tandon School of Engineering**

Master of Science in Telecommunication Networks
Thesis - CitySynth: Imaging with a Network of Devices

Brooklyn, NY, USA
 Aug. 2012 - May. 2014

SELECTED PROJECTS

- **AdguardController:** Developed a Kubernetes controller to automate DNS record management, replacing manual infrastructure changes with a declarative control loop.
- **UCSLHUB:** Developed resilient backend infrastructure using Kubernetes, JupyterHub, and Keycloak for the CUSP UCSSL bootcamp, serving hundreds of students each year.
- **Homelab:** Operate a distributed HCI lab with Ansible and Terraform-managed KVM nodes, Ceph storage, Kubernetes, and network segmentation for testing distributed systems and multi-datacenter failure scenarios.

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